

LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

Generate a Research Paper with AI, Then Audit Its Citations

AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40

Student Detail	Fill in
Name	Subhadip Bag
Roll Number	MRA2026005
Programme / Department	MTech IT(Robotics & AI)
Topic ID	T 11
Full Assigned Topic	Single-LLM Versus Multi-Agent Debate Systems for Scientific Hypothesis Generation
AI Tool Used	Claude
Date	21/08/26

IMPORTANT: You will receive a topic from T1-T40. Your task is not to produce the “best” paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

Question 1 - Citation Authenticity: Does the cited publication actually exist, and is it correctly identified?

Question 2 - Claim-Citation Support: If the publication is genuine, does it actually support the claim for which the AI cited it?

2. IMPORTANT RULES

- Use only the topic assigned to you by the instructor.
- Generate the paper in a fresh AI conversation.
- Do not tell the AI that you are conducting a citation audit or hallucination test.
- Do not intentionally ask the AI to generate fake references.
- Preserve the original AI-generated paper exactly as produced.
- Do not correct, regenerate, replace, or question references before saving the original output.
- For verification, do not accept another AI chatbot's statement as proof that a reference exists.
- Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources.
- Record what you actually find, even if all references are genuine or all are problematic.

Most important principle: You are not being evaluated on whether the AI produces good or bad references. You are being evaluated on how systematically and correctly you audit them.

3. WHAT YOU WILL DO

Part	Activity
A	Record your baseline perception of AI citations
B	Record the AI environment used
C	Generate and preserve an AI-written research paper
D	Create a reference inventory
E	Select 10 references systematically
F	Judge their plausibility before verification
G	Verify authenticity, metadata and identifiers
H	Determine whether genuine references support the AI-generated claims

4. RECOMMENDED 2-HOUR TIME PLAN

Time	Activity
0-10 min	Parts A-B: Baseline and AI environment
10-30 min	Part C: Generate and save research paper
30-40 min	Parts D-E: Inventory and select references
40-45 min	Part F: Pre-verification plausibility
45-80 min	Part G: Authenticity and metadata audit
80-110 min	Part H: Claim-citation entailment audit
110-120 min	Calculate scores, check worksheet and prepare submission

Do not spend excessive time formatting the AI-generated paper. The audit is the main activity.

PART A - BASELINE ASSESSMENT

Complete this section BEFORE opening the AI tool.

A1. How often do you use generative AI for academic work?

☐ Never ☐ Rarely ☐ Occasionally ☐ Weekly ☒ Almost daily

A2. Have you previously asked an AI system to generate references?

☐ Frequently ☒ Occasionally ☐ Once or twice ☐ Never

A3. Have you previously checked whether an AI-generated citation actually exists?

☐ Always ☐ Sometimes ☒ Rarely ☐ Never ☐ Not applicable

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5 (1 = not at all confident; 5 = completely confident)

Important: Do not change this answer later, even if your opinion changes during the experiment.

PART B - RECORD YOUR AI ENVIRONMENT

Information	Your Response
-------------	---------------

AI tool used	Claude
Model name	Sonet
Model/version shown	Sonet 5
Access tier	☒ Free ☐ Paid/Pro ☐ Institutional ☐ Not sure
Web browsing available?	☐ Yes ☐ No ☒ Not sure
Was web browsing actually used?	☒ Yes ☐ No ☐ Not sure
Research/Deep Research mode used?	☒ Yes ☐ No
Date/time generated	21/08/26

Important: If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

PART C - GENERATE THE AI RESEARCH PAPER

C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

C2. Use the Following Common Prompt

Replace only [YOUR ASSIGNED TOPIC].

Generate a scholarly research paper on the topic: "[YOUR ASSIGNED TOPIC]". The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.

C3. Preserve the Original Output

- ☐ Save the complete AI-generated paper
- ☐ Save the complete reference list
- ☐ Take screenshot(s) showing the generated output/references
- ☐ Save conversation/export/share evidence if available

<https://claude.ai/share/b99e4ace-fe9e-4e22-beca-a505b7476b6b>

STOP: Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title: Single-LLM Versus Multi-Agent Debate Systems for Scientific Hypothesis Generation

Measure	Value
---------	-------

Approximate pages	8
Approximate word count	5572
Number of sections	10
Total number of references	24
Approximate in-text citation occurrences	66
Did AI warn you to verify references?	☐ Clearly ☐ Vaguely ☒ No

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	4
PMID available, no DOI	0
arXiv ID available, no DOI/PMID	18
URL only	1
No persistent identifier supplied	1
TOTAL	24

Examples

Category	Example	How to Count
DOI available	Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730	DOI available
PMID available, no DOI	A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found	PMID available, no DOI
arXiv ID available	Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629	arXiv ID available
URL only	World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID	URL only
No persistent identifier	Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.	No persistent identifier if none is supplied

Important: Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

If the bibliography contains 10 or fewer references: audit all references.

If it contains more than 10 references: select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

Audit Slot	Original Bibliography Reference No.
R1	1

R2	2
R3	3
R4	7
R5	11
R6	15
R7	19
R8	22
R9	23
R10	24

Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this **BEFORE** searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: "If I had not verified this citation, how convincing would it look to me?"

Score	Meaning
1	Obviously suspicious
2	Somewhat suspicious
3	Unsure
4	Looks genuine
5	Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
R1	Full 8-author list, specific arXiv ID	5	Yes / No
R2	Clean 3-author list, venue tag adds specificity	5	Yes / No
R3	Only one named author + "et al." with no co-author names given	3	Yes / No
R4	DOI present, matches known Nature/arXiv dual-publication pattern	5	Yes / No
R5	Short, clean, recognizable author trio	5	Yes / No
R6	No author names at all — just a title and journal	2	Yes / No
R7	Well-known author trio (Stanford NLP), specific ID	5	Yes / No
R8	Canonical, widely-cited paper, correct author ordering	5	Yes / No
R9	Plausible but 2026 date + arXiv ID prefix "2604" (April 2026) is unusual enough to raise a small flag for anyone unfamiliar with post-cutoff search-sourced material	4	Yes / No

R10	DOI present, specific journal (Journal of King Saud University), matches a real Springer publication pattern	5	Yes / No
-----	--	---	-----------------

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Never change Part F after completing Part G. Your incorrect predictions are valuable experimental observations.

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist?
- Is the title correct?
- Are the authors correct?
- Is the year correct?
- Is the journal/conference/source correct?
- Are volume, issue and pages correct, if supplied?
- Does the DOI/arXiv/PMID belong to this publication?

G2. Verification Procedure

- ☐ Persistent identifier: DOI / arXiv / PMID
- ☐ Publisher or official repository
- ☐ Google Scholar
- ☐ Semantic Scholar / OpenAlex
- ☐ Crossref
- ☐ PubMed or another discipline-specific database
- ☐ Exact-title search
- ☐ Author + title-keyword search

Not acceptable as sole evidence: “I asked another AI and it said the paper exists.” AI can help you search, but final verification must rely on scholarly or publisher evidence.

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata are correct.
B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.

D	FABRICATED	No reliable scholarly evidence that the publication exists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G4. Examples

A: AI gives a paper and all details match the publisher record.

B: The paper exists, but the AI gives the wrong publication year or journal.

C: The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

D: You cannot find the paper or a credible scholarly record after systematic searching.

E: The DOI supplied by AI opens an entirely different paper or does not resolve.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?
R1	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R2	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R3	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R4	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R5	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R6	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R7	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R8	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R9	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA
R10	Y/N	Y/N	Y/N	Y/N	Y/N	Y/N/NA

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1	https://arxiv.org/abs/2308.07201	arXiv:2308.07201	Exact match: authors, title, ID all correct
R2	https://arxiv.org/abs/2508.17536	arXiv:2508.17536	Exact match; NeurIPS 2025 Spotlight status confirmed independently
R3	https://arxiv.org/abs/2509.05396	arXiv:2509.05396	Real authors are Andrea Wynn, Harsh Satija, Gillian Hadfield — "Doshi" does not appear anywhere in the record
R4	https://www.nature.com/articles/s41586-026-10652-y	arXiv:2505.13400	Full 10-author list matches exactly
R5	https://arxiv.org/abs/1805.00899	arXiv:1805.00899	Matches
R6	https://www.cell.com/science/fulltext/S2589-0042(25)02245-X	None captured	Paper genuinely exists (Coated-LLM, iScience) but I never recovered author names in any search result and cited it with none — and no DOI was captured
R7	https://arxiv.org/abs/2409.04109	arXiv:2409.04109	Matches
R8	https://arxiv.org/abs/2201.11903	arXiv:2201.11903	Matches

R9	https://arxiv.org/abs/2604.24881	arXiv:2604.24881	Matches
R10	https://doi.org/10.1007/s44443-025-00353-3	DOI:10.1007/s44443-025-00353-3	First author "Yan Zhou" confirmed; full co-author list beyond "et al." not independently verified

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	A/B/C/D/E	Publication exists; authors, title, arXiv ID all verified correct.
R2	A/B/C/D/E	Publication exists; authors, venue (NeurIPS 2025 Spotlight), and arXiv ID all verified correct.
R3	A/B/C/D/E	Article exists and arXiv ID is correct, but I attributed it to "Doshi, A." when the actual authors are Wynn, Satija, and Hadfield.
R4	A/B/C/D/E	Publication exists; full 10-author list and DOI verified correct.
R5	A/B/C/D/E	Canonical paper; authors, year, and arXiv ID verified correct.
R6	A/B/C/D/E	Article exists in iScience, but I supplied no author names and no identifier, leaving the citation materially incomplete.
R7	A/B/C/D/E	Publication exists; authors, arXiv ID, and study details verified correct.
R8	A/B/C/D/E	Canonical NeurIPS 2022 paper; authors and arXiv ID verified correct.
R9	A/B/C/D/E	Publication exists; authors and arXiv ID verified correct against the source PDF.
R10	A/B/C/D/E	Publication exists; DOI and first author verified via independent press coverage, though full co-author list unconfirmed.

Good reason: B: Article exists, but AI reported 2022 instead of 2020.

Poor reason: B: Details wrong.

PART G - SCORING

Count your final classifications: A = 8, B = 2, C = 0, D = 0, E = 0

Total audited: N = A + B + C + D + E = 10

Code	Points
A	4
B	3
C	1
D	0
E	1

Authenticity Points = $4A + 3B + C + E$

Authenticity Score = $[(4A + 3B + C + E) / (4N)] \times 100$

My calculation: $4(8) + 3(2) + 1(0) + 0(0) + 1(0) = 38$ points

Authenticity Score = $38 / (4 \times 10) \times 100 = 95.0 / 100$

Worked example: If A=6, B=2, C=1, D=1, E=0, then points = $4(6)+3(2)+1(1)=31$; maximum = 40; Authenticity Score = $31/40 \times 100 = 77.5$.

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns
60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk
Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = $(\text{Correct genuine/fake predictions} / \text{Total predictions}) \times 100$

Correct predictions = 10, Total predictions = 10, Prediction Accuracy = 100 %

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	24
References deeply audited	10
A: Verified	8
B: Wrong metadata	2
C: Frankenstein	0
D: Fabricated	0
E: Identifier mismatch	0
Authenticity Score	95 / 100
Prediction Accuracy	100 %

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

Ans: Professional-looking citations weren't always genuine in every detail - all 10 papers existed, but 2 had wrong or missing metadata (a fabricated author name in one, no authors at all in another) despite looking equally polished as the correct ones..

2. Did all genuine references actually support the claims for which they were cited?

Ans: This lab checked existence and metadata, not claim-support - I didn't verify whether each citation's content actually backs the specific claim it was attached to, which is a separate, unaudited risk.

3. What was the most serious citation-integrity problem you observed?

Ans: The most serious issue was the fabricated author name ("Doshi, A." for a paper actually written by Wynn, Satija, and Hadfield) - an active misattribution, not just an omission, and one I had no signal was wrong until forced to check.

4. What is the single most important lesson you learned from this lab?

Ans: Key lesson: verify each bibliographic field individually - "the paper is real" and "the citation is accurate" are different checks, and fluency/formatting is never evidence of correctness.

SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

Example: Lab1_IEC2026001_RahulSharma

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Student Name:Subhadip Bag.....

Roll Number:MRA2026005.....

Signature: ...Subhadip Bag.....

Date:21/0812026.....

CENTRAL LESSON OF LAB 1

Does the source exist? + Does the source actually support the claim?

AI Generation -> Reference Inventory -> Systematic Sampling -> Prediction -> Authenticity Verification -> Claim-Support Verification -> Quantitative Scores -> Submission